REFLOWECONOMY · RISK REGISTER AND HAZARD ANALYSIS

ReflowEconomy risk register and hazard analysis

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DOCUMENT CONTROL

Document ID	RFE-RSK-001	Version	0.1
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/refloweconomy		
Source	docs/09-risk.md @ 9819cf9		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First risk register, preliminary hazard analysis and first-pass design FMEA (kit 1.8 pipeline, stage P3) for the reference micro-factory (option A)

ReflowEconomy risk register and hazard analysis

The top project risks are the electrical supply (the morning loads exceed a 40 A single-phase supply once motor currents are counted), product prices too low to pay for the equipment, and the input mix carrying more residue than the 10 % the disposal target needs. The top hazards are the shredder, melt fumes, hot platens, lithium cells in the input and the fire load of plastic stock. Method: STANDARDS section 23. Likelihood, impact, severity, occurrence and detection are scored 1 (low) to 5 (high); for detection, 5 means the failure is hardest to detect.

> **Safety:** This is a paper analysis at TRL 3. A qualified safety professional, a licensed electrician and, for the container option, a structural engineer must review the design before any site operates (RFE-BLD-001 stops S1 to S7).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	The site cannot run on the supply it can get	Motor drives and single-phase motors draw about 55 A in the morning block against a 40 A single-phase supply (RFE-STM-001 S14)	Breakers trip; or the line runs fewer hours or less input	5	4	Design lead, with Amish	Decision D1: three-phase recommended; single-phase fallback in three blocks
P2	Products do not sell at the assumed price	The site needs 2.49 USD/kg; regenerated HDPE sheet is listed in India at about 0.5 to 1 USD/kg (RFE-MKT-001)	Losses every shift	4	5	Partner and operator	Design-grade sheets and furniture for named buyers; staged start as a sort-and-bale hub (RFE-DEC-001 D4)
P3	Input carries too much residue	Mixed collection; the 10 % intake threshold refuses loads pickers bring	Disposal over 20 % (PR-008) or loss of supply goodwill	4	3	Partner	Agree the intake rule with collectors in co-design (UR-019)
P4	Machine rates below the assumptions	Shredder at the low end of 8.8 to 41.8 kg/h; derated press	Less than 100 kg a shift (PR-006)	3	3	Design lead	Measure at TRL 4; three-phase supply allows the published 15 kW press
P5	No co-design partner agrees	Partner priorities and capacity	Requirements stay first-pass; no site	3	4	Amish	Approach SWaCH through WIEGO as decided; second candidates in the plan
P6	Permits take long or cost more	Consent route and category not yet known (RFE-REG-001 G2)	Delay before operation	3	3	Operator with a consultant	Assess the category early with MPCB
P7	Exposure limits exceeded in practice	Hood gaps, sash left open, filters not changed	Harm to workers; site closure	2	5	Safety professional	Face velocity checks, filter log, air monitoring at TRL 4

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Shredder cutters	Hand or arm reaching through the feed or the door	Amputation	5	2	Cranked feed hood, 120 mm slot and 1,060 mm reach path (ISO 13857 Table 4), interlocked flap and door, emergency stops, lockout	Low
H2	Melt fumes	Formaldehyde, acrolein and aerosol from HDPE and PP at 190 to 210 C; PVC releasing HCl if missed at sorting	Respiratory harm	4	3	Two enclosing hoods at 0.5 m/s, sash covering one opening, filters and stack, PVC excluded at sorting	Medium until air monitoring
H3	Hot platens and barrel	Contact at 190 to 200 C	Burns	4	3	Insulated platen packs, booth with sash, gloves and face shields, cool-down rule	Medium
H4	Press closing force	Hands between platens	Crushing	4	2	Mould loaded on the transfer bridge, hands clear before the jacks are pumped; sash closed while pressing	Low
H5	Noise	Shredder in a reverberant steel room	Hearing loss	3	3	Lined enclosure (79 dB(A) on paper), hearing protection zone until measured	Medium
H6	Electric shock	Water near electricity at the tanks	Death or injury	5	2	30 mA RCDs, splash-rated sockets, licensed installation	Low
H7	Lithium cell fire	A cell in the input reaches the shredder or is crushed	Fire, smoke	5	3	Cells pulled out at intake, dry-sand container in the cage, three extinguishers	Medium
H8	Fire in plastic stock	Ignition near flake and products; dust	Fire	5	2	Stock 1.0 m from the hot zone, daily cleanup, exits at both ends	Medium
H9	Sharps and contaminated input	Needles, glass, chemical containers	Cuts, infection, poisoning	3	4	Puncture-resistant gloves, sharps container, refusal of medical and chemical waste	Medium
H10		Bags and sheets	Back injury	3	3		Low

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
	Manual handling					Bags of 25 kg or less; transfer bridge between presses	
H11	Container structure	Side walls removed at the seam	Collapse	5	1	Seam beam and propping to a structural engineer's design; shed preferred	Low
H12	Low headroom	Cable tray crossing at 2.04 m	Head strike	2	2	Padding and hazard tape (2.02 m under the padding)	Low

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Electrical supply	Carry the morning loads	Breaker trips under load	Line stops; heaters cool mid-cycle	Input currents about 55 A on a 40 A supply	4	5	2	40	Decision D1; clamp-meter check at TRL 4 (BLD first checks)
F2	Heater interlock	Keep the press heaters off while a motor runs	Contactor welded or bypassed	Overload, nuisance trips, overheating cables	Contact wear; manual bypass	4	2	4	32	Set a performance level (ISO 13849-1); test at every start-up
F3	Feed flap interlock	Stop the rotor when the flap opens	Switch defeated or fails to danger	Exposure to the cutters	Tampering; single-channel switch	5	2	3	30	Coded or guard-locking switch (ISO 14119); daily check
F4	Feed hood and slot	Keep a hand 850 mm from the cutters	Slot widened or hood removed for big parts	Reach to the cutters	Operators feeding oversize pieces	5	2	3	30	Pre-cut and crush at sorting; bolted hood; training
F5	Fume fan and filters	Hold 0.5 m/s at the hood faces	Face velocity drops	Fume exposure	Filters loaded; damper moved; fan belt or motor fault	4	3	4	48	Filter log; pressure gauge across the filters; anemometer check monthly
F5a	Fume duct	Carry fume to the filters	Wax condensate builds up and catches fire	Duct fire	Aerosol condensing in the cool galvanised duct	5	2	4	40	Cleaning schedule and inspection hatch; no heat source downstream of the hoods
F6	Booth sash	Leave only one opening uncovered	Both openings open	Lower face velocity	Sash removed for cleaning and not refitted	3	2	2	12	Stops on the rails; check before heaters on (stop S4)
F7	Shredder enclosure	Cut noise by about 15 dB	Insertion loss well below 15 dB	Exposure above 85 dB(A)	Gaps, door seal worn, slot flap left open	3	3	3	27	Measure at TRL 4; seal and flap checks
F8	Enclosure ventilation	Keep the gearmotor cool	Motor overheats	Trip or motor damage	No vent in the lined enclosure	3	3	3	27	Confirm temperature when bought (DEC confirm item 3); lined vent baffle if hot
F9	Float-sink tank	Separate PP and HDPE			Filled or foamed plastics; PVC not	4	3	3	36	Sorting rule; NIR or WasteWise

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
		from PET and PVC	PVC or filled plastics pass to the melt	HCl fumes; product defects	removed at sorting					check for unclear parts
F10	Hot press	Hold 190 to 200 C through the cycle	Platen overheats past 230 C	Degraded melt, more fume	Thermocouple fault or controller failure	4	2	3	24	Independent over-temperature cut-out
F11	Cell container	Hold a burning cell	Fire spreads	Fire in the cage	Sand missing or container left open	5	2	3	30	Lid and sand checked at each shift; extinguisher at the entrance
F12	Seam beam (container option)	Carry the roof edges over the 12 m seam	Lateral-torsional buckling	Roof sags or fails	Top flange not tied to the roof	5	1	4	20	Structural engineer's design ties the flange (RFE-STM-001 S18)
F13	Cable tray crossing	Cross the aisle above head height	Head strike	Minor injury	Padding missing	2	2	2	8	Padding and tape in place (stop S7)
F14	Passport desk	Issue a valid passport for every lot	Lot leaves without a passport	Buyer distrust; export refused	Rush at the end of a shift	3	3	3	27	Labels printed only from a valid record; end-of-shift check